

WHAT IS IT

This document defines the complete project creation flow in NuBred — from the first document upload to a fully confirmed and active project. It specifies what the AI extracts from each document type, how confidence levels are assigned, how conflicts are resolved, and for every section of the platform: exactly what NuBred extracts, from which sources, with which confidence level, and what the VM must do to confirm it.

V2 replaces V1 entirely. Three major additions: (1) Input Channels — Manual Upload, Connected Sources, Paste Text. (2) Confidence Level Framework — Confirmed, Provisional, Conflicting. (3) Per-section analysis for every module including Quarantine, Protocols, IP Rights, and Chronology.

THE CORE PRINCIPLE

The VM never fills in forms from scratch. NuBred does the work. The VM confirms, corrects, and improves. Every piece of information in NuBred has a cited source. Every extracted field has a confidence level. The VM is always in control of what becomes definitive.

Documents can be uploaded at any time — before and after project activation. Each new upload triggers a new AI analysis cycle. The project record improves continuously as more documents are added. There is no final state — a project is always open to new information.

INPUT CHANNELS — THREE WAYS TO BRING INFORMATION INTO NUBRED

NuBred accepts information from three channels. All three feed the same AI pipeline. The difference is the source and the default confidence level assigned to extracted data.

Channel	How it works	Default confidence	VM action required
Manual Upload	VM drags and drops files directly. Any format accepted: PDF, DOCX, Excel, email export, audio transcript. VM can tag the document type at upload.	Depends on document type. Signed contract → Confirmed. Email export → Provisional. VM can override at upload.	Upload + optional type tag. NuBred analyzes immediately.
Connected Sources	NuBred connects to Outlook, Gmail, or Google Drive. Monitors for content relevant to active projects. Presents items in a review queue — VM decides what to import.	Always Provisional on import. VM upgrades to Confirmed when they verify the content is definitive.	VM reviews queue, approves import, assigns document type, confirms or discards.
Paste Text	VM pastes raw text — WhatsApp messages, chat exports, text from external portals. NuBred treats as unstructured document and extracts what it can.	Always Provisional. VM must confirm any extracted data before it becomes active.	Paste + confirm. VM sees extraction and explicitly marks what is reliable.

Connected Sources require explicit VM approval for every item before it enters the project. NuBred never silently imports emails or documents from a connected mailbox. The VM always sees a review queue and decides what is relevant and what to discard.

CONFIDENCE LEVEL FRAMEWORK

Every extracted field in NuBred has a confidence level. This is the system's way of communicating how certain it is that the data is correct and definitive. The VM can override any confidence level — but the system always shows the original assessment.

Level	Definition	Typical source	UI indicator
Confirmed	Data extracted from a signed contract, official certificate, or source explicitly marked as definitive by the VM.	Signed PDF, PBR certificate, document marked as Final by VM, NuBred standard framework rules.	Green badge. No warning shown.
Provisional	Data extracted from a non-binding source. May be superseded by a later signed document.	Email, meeting notes, WhatsApp transcript, draft contract, paste text, unverified oral agreement.	Orange badge. Tooltip: 'Source: [email/meeting notes]. Not yet confirmed by a signed document.'
Conflicting	NuBred found different values for the same field in two or more sources. VM must resolve before confirming.	Any scenario where two documents state different values for the same field — e.g. royalty 8% in email, 7% in draft contract.	Red badge. Confirm button blocked until VM resolves. Side-by-side comparison shown.

When a Connected Sources email contains a royalty figure, NuBred marks it Provisional and shows the VM: 'Royalty 8% — extracted from email dated 3 March. This has not been confirmed in a signed contract.' The VM confirms the project knowing this data point is not yet final.

CONFLICT RESOLUTION

Conflicts occur when two documents state different values for the same field. NuBred detects conflicts automatically and blocks confirmation of the affected section until the VM resolves them.

Scenario	How NuBred handles it
Same field, two different values in two documents	Conflicting badge on the field. Side-by-side comparison showing both values with their cited sources. VM selects which value to keep. The discarded value is archived with its source — never deleted.
Signed contract overrides an earlier email value	NuBred detects that the signed contract was executed after the email and auto-upgrades the contract value to Confirmed. Flags the email value as superseded. VM sees the resolution and confirms.
Negotiation still open — no signed version yet	Both values remain Provisional. NuBred cannot determine which is correct. VM must manually mark one as the working value and add a note explaining the status. Confirmed badge only available after a signed document arrives.
VM overrides AI extraction	VM can manually correct any extracted field at any time. The override is logged with timestamp, user, and the original AI extraction preserved as reference. This creates an audit trail.

THE FUNDAMENTAL FLOW

Project creation follows four steps in a loop that can repeat indefinitely.

#	Step	What happens	Who acts
1	Bring in documents	VM uploads files, connects Outlook/Gmail/Drive, or pastes text. Any format, any language, any number of documents.	VM. NuBred accepts and queues for analysis.
2	AI Analysis	Claude applies three layers simultaneously: NuBred Glossary, Phase & Protocol Framework, AgreeLyze Clause Library. Assigns confidence levels to every extracted field.	Automatic. VM sees progress by section, not a spinner.
3	Section-by-section confirmation	VM reviews each section in sequence. Sees AI extraction + original document side by side. Confirms, edits, uploads more documents. Each section has its own AI chat.	VM. No section is auto-confirmed.
4	Iterative improvement	New documents can be added at any time — before or after project activation. Each addition triggers a new analysis cycle for affected sections only.	VM. Continuous loop.

PER-SECTION ANALYSIS — MVP SECTIONS

The following analysis covers every section of NuBred in detail. For each section: what the AI extracts, from which sources, confidence levels, how the VM confirms, and what blocks confirmation.

SECTION 1 — OVERVIEW

Overview — AI Extraction & Confirmation

What AI extracts	Governance Model type (Nursery-Led, Club Model, Open Market, etc.), 13-parameter rating, project name, species, territory, start date, number of actors, primary commercial objective.
From which sources	All uploaded contracts analyzed together. The Governance Model is inferred from the combination of contract types, actor relationships, and royalty structure — not from a single document.
Confidence level	Always Provisional on first generation. The Governance Model is a classification produced by NuBred — it requires explicit VM confirmation before it becomes the active project type.
How VM confirms	VM sees the Governance Model card with the 13-parameter rating. Can adjust individual ratings, change the model type, or add a note explaining the project context. Confirm button available immediately — no blocking fields.
What blocks confirmation	Nothing blocks Overview confirmation. But VM must have viewed Overview before the project can be activated. It is the first section in the confirmation sequence.
Connected Sources note	If Outlook is connected, NuBred may identify the project name and territory from email subject lines and signatures. These are always Provisional and appear as suggestions, not pre-filled values.

SECTION 2 — GENOTYPE

Genotype — AI Extraction & Confirmation

What AI extracts	Species (botanical name and common name), variety or selection names, internal breeder codes, number of plants per genotype, development status (Selection = no IP protection, Variety = protected). Links genotypes to phases and IP records.
From which sources	Contracts (variety name in preamble and annexes), protocols (genotype list and codes), planting plans (number of plants per variety), IP certificates (denomination and species), emails (variety names mentioned in coordination).
Confidence level	Variety name from signed contract = Confirmed. Variety code from protocol = Confirmed. Number of plants from email = Provisional. Status (Selection vs Variety) = Confirmed if PBR certificate uploaded, Provisional otherwise.
Conflicting scenario	Most common conflict: same variety referred to with different names in different documents (breeder code in contract, commercial name in protocol). NuBred flags as Conflicting and shows both versions. VM confirms the canonical name.
How VM confirms	VM reviews the genotype list. Can rename, merge duplicates, add missing genotypes manually, link each genotype to its IP record. Confirm available once all genotypes have at least a name and species.
What blocks confirmation	Any genotype without a species assigned blocks confirmation. Species is the minimum required field.

SECTION 3 — SUPPLY CHAIN

Supply Chain — AI Extraction & Confirmation

What AI extracts	All actors with roles (Variety Manager, Nursery, Grower, Packhouse, Marketer), relationships between actors, the contract governing each relationship, and edge types (Propagation, Fruit Sales, Licensing, Reporting, Unconfirmed).
From which sources	All contracts (parties identified from preambles and signatures), emails (actors mentioned by name or company), planting plans (grower names and locations), IP certificates (filing company = Breeder/VM).
Confidence level	Actor identified in signed contract = Confirmed node. Relationship governed by signed contract = Confirmed edge. Actor mentioned only in email = Provisional node. Relationship inferred from email without contract = Unconfirmed edge (dashed orange in diagram).
Conflicting scenario	Same company appears with different roles in different documents (e.g. Grower in contract, Packhouse in email). NuBred flags as Conflicting. VM assigns the correct role for this project.
How VM confirms	VM reviews the actor diagram. Resolves Unknown Actor nodes (role unclear), confirms or corrects Unconfirmed edges, adds missing actors manually. Dragging nodes repositions them in the diagram without affecting data.
What blocks confirmation	Any Unknown Actor node (role not determined) blocks Supply Chain confirmation. Any Unconfirmed edge that references a missing contract triggers a warning — VM can confirm the relationship manually without uploading the contract, but must explicitly acknowledge the absence.
Connected Sources note	With Outlook connected, NuBred identifies new actors from email signatures and CC lists. These always appear as Provisional nodes in a review queue before entering the diagram.

SECTION 4 — PHASES

Phases — AI Extraction & Confirmation

What AI extracts	Phase names, types (Trial/Pilot/Launch/Scale), objectives, duration, location, number of plants or hectares per phase, gate criteria, genotype decisions (Promote/Repeat/Discard) if documented.
From which sources	Contracts (phase definitions in body or annexes), protocols (phase-specific observation requirements), meeting notes (phase decisions), emails (phase advancement communications).
Confidence level	Phase defined in signed contract = Confirmed. Phase duration from email = Provisional. Gate criteria from protocol = Confirmed if protocol is official, Provisional if it is a draft.
Glossary mapping	NuBred uses the standard glossary to map non-standard terminology. 'Evaluation period' → Trial. 'Commercial introduction' → Launch. 'Preliminary testing' → Trial. Every mapping is shown to VM with cited source via the Why Button.
Conflicting scenario	Contract states 2-year Trial, email states 3-year Trial. NuBred flags Conflicting. VM resolves — typically the signed contract takes precedence, but VM confirms.
How VM confirms	VM reviews the phase timeline. Can rename phases, adjust types, modify dates and durations, add missing phases, set gate criteria. Must assign at least one phase before confirming.
What blocks confirmation	At least one active phase is required. A project with no phases identified cannot be confirmed.

SECTION 5 — CONTRACTS

Contracts — AI Extraction & Confirmation

What AI extracts	Contract type (AgreeLyze Economic Function code MF-1 to MF-8), parties, clauses present/absent with risk flags, gap analysis (ABSENCE/ANOMALY/IMBALANCE), obligations per party, royalty structure, territorial scope, governing law, termination triggers.
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From which sources	All uploaded contract files. NuBred processes each contract independently and then cross-references them to identify consistency gaps across the contract suite.
Confidence level	Uploaded signed contract = Confirmed source. Clause analysis findings = always Provisional until VM confirms. A clause flagged as HIGH risk is not a confirmed problem — it is a finding that requires VM review.
Clause analysis detail	For every HIGH and MEDIUM risk clause, and every absent clause, NuBred generates: what the clause says in this contract, what is missing if absent, what imbalance exists if detected, and a short recommendation. Every analysis cites the exact passage.
How VM confirms	3-tab review screen: Tab 1 Actors diagram, Tab 2 Clause analysis, Tab 3 Gaps. VM must view the Gaps tab before the Confirm button is active. VM can add notes per clause. Notes are stored in the project record.
What blocks confirmation	Missing information fields flagged by NuBred block confirmation. VM must fill them manually. Gaps tab must be viewed. No clause finding itself blocks confirmation — the VM decides what is acceptable.
Connected Sources note	With Outlook connected, draft contracts arriving by email are imported as Provisional. When the signed version arrives, NuBred compares it to the draft and flags any changes between the two versions for VM review.

SECTION 6 — IP RIGHTS

IP Rights — AI Extraction & Confirmation

What AI extracts	IP type (PBR, Patent, Trademark), registration number, filing company, territorial scope, registration date, expiry date, grant number, variety denomination, species, linked genotype. For Trademark: type (word/figurative), Nice class, associated IP rights.
From which sources	PBR/patent certificates (primary source), contract annexes with IP schedules, licensing contracts with IP clause references, emails mentioning registration numbers.
Confidence level	Data from official certificate = Confirmed. Data from contract reference = Confirmed if contract is signed. Registration number from email = Provisional. Expiry date inferred from registration date + standard duration = Provisional — must be confirmed by certificate.
Deduplication logic	Before creating a new IP record, NuBred checks for existing records with the same registration number. If found, it updates the existing record rather than creating a duplicate. VM is notified of the update.
Automatic alerts generated	180-day expiry warning → To-Do for VM. Territory gap (grower in country not covered by IP) → Notification + To-Do. Missing trademark when Brand Royalty is active → Notification. PBR exists but no associated Trademark → Warning flag on IP record.
How VM confirms	VM reviews IP cards grouped by type (PBR first, then Patent, then Trademark). Links IP records to genotypes and to other IP records (PBR ↔ TM). Confirm available once all IP records have at least type, registration number, and territory.
What blocks confirmation	IP record with no registration number blocks confirmation. IP record with no territorial scope blocks confirmation. These are the minimum required fields for governance integrity.

SECTION 7 — TO-DO

To-Do — AI Generation & Confirmation

What AI generates	Task list per actor organized by season and type. Each task has: title, due date, category (reporting/payment/observation/delivery/certification), assigned actor, and cited source.
Two source types	Source 1: NuBred standard framework — obligations every actor of that type has in that phase (e.g. every Grower in commercial phase must declare harvest actuals within 30 days of harvest). Source 2: Contract obligations — deadlines and duties extracted from uploaded contracts.

Confidence level	Framework-generated tasks = Confirmed (they are NuBred rules, not extractions). Contract-extracted tasks from signed contracts = Confirmed. Contract-extracted tasks from draft/email = Provisional. Due dates from emails = Provisional.
Conflicting scenario	Two contracts assign the same obligation to different parties. NuBred flags Conflicting assignment. VM resolves by confirming which party is responsible.
How VM confirms	VM reviews the full task list. Can edit titles, adjust dates, reassign tasks, add custom tasks, and delete tasks that are not applicable. Confirm available once the list has been reviewed.
What blocks confirmation	Nothing blocks To-Do confirmation — the VM can confirm an empty or partial list. However, any Conflicting task assignment must be resolved before the section can be marked as reviewed.
Connected Sources note	With Outlook connected, NuBred monitors for emails that contain deadline confirmations or obligation reminders. These are added to the review queue as Provisional To-Do suggestions.

PER-SECTION ANALYSIS — OUT OF MVP TAB BAR

These sections are not visible in the MVP tab bar but their data models are active. The AI extraction logic below applies from day one.

PROTOCOLS & OBSERVATIONS

Protocols — AI Extraction & Confirmation

What AI extracts	Protocol name, objective, phase linkage, parameter family (F1-F7), observation parameters with input type and measurement method, frequency, data collection windows, sample sizes, quality thresholds, eliminatory criteria.
From which sources	Technical protocol documents (primary source), contract annexes with protocol references, VCU evaluation guidelines, trial agreement schedules.
Confidence level	Parameters from official protocol document = Confirmed. Frequency from contract = Confirmed. Threshold from email discussion = Provisional. Custom parameters added by VM = Confirmed (VM is the source).
Glossary mapping	NuBred maps non-standard protocol terminology to the standard parameter family schema. 'Soluble solids content' → Brix (F2). 'Tree cross-section area' → Tree Vigour (F1). Every mapping shown via Why Button.
Field Observer assignment	Field Observer assignments are not extracted from documents — they are assigned by VM after protocol confirmation. NuBred generates the assignment interface based on the extracted protocol structure.
How VM confirms	VM reviews extracted parameters, adjusts frequencies and thresholds within allowed ranges, adds custom parameters, assigns Field Observers to genotypes. Minimum requirement: at least one parameter per active protocol.
NuBred Node note	The GET /api/protocols/active endpoint is populated from confirmed protocol data. NuBred Node calls this endpoint to receive its instructions. The data schema is identical for manual Field Observer and hardware Node.

QUARANTINE

Quarantine — AI Extraction & Confirmation

What AI extracts	Location (country and facility), quarantine type (Public/Private), estimated duration, start date, expected certification date, linked genotypes, reference contract or nursery agreement.
From which sources	Phytosanitary certificates (primary source), nursery supply agreements with quarantine clauses, import permits, emails from quarantine facilities with status updates.
Confidence level	Data from official certificate = Confirmed. Duration from contract clause = Confirmed. Expected certification date from email = Provisional. Quarantine requirement inferred from contract without explicit clause = Provisional.
Status computation	Status is never extracted — it is computed by NuBred from the data: In Progress, Awaiting Certification (30 days before expected cert date), Complete (certificate uploaded), Overdue (past expected date, no certificate). Computed on every read.
How VM confirms	VM reviews quarantine records, confirms location and duration, links genotypes, uploads certification documents when received. When a certificate is uploaded, status automatically updates to Complete.
What blocks confirmation	Quarantine record with no location blocks confirmation. Quarantine record with no linked genotype blocks confirmation.

Connected Sources note	With Outlook connected, NuBred monitors for emails from quarantine facilities. Status updates and certification notifications are added to the review queue automatically.
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CHRONOLOGY (ROADMAP — DATA MODEL ACTIVE)

Chronology — AI Extraction Logic	
What AI extracts	All events with explicit or inferable dates across all uploaded documents: contract signatures, email exchanges, meeting decisions, phase advancements, observation milestones, IP filings, quarantine certifications, harvest declarations.
From which sources	Every document type. NuBred extracts dates from document metadata, body text, email headers, and explicit date references ('on 15 March 2024', 'effective from Q1 2025').
Confidence level	Date from document metadata = Confirmed. Explicit date in signed contract = Confirmed. Date mentioned in email body = Provisional. Inferred date ('in the first year of the Trial phase') = Provisional with estimated range.
How events are ordered	Chronological order by event date. Where dates conflict, the most recent confirmed source takes precedence. VM can manually reorder events or correct dates.
UI when live	Vertical timeline with event cards. Each card shows: date, event description, document source (clickable Why Button), confidence badge. Events group by month. VM can filter by type or actor.
What blocks implementation	Nothing blocks the data model — chronology events are generated as a side effect of all other section analyses. The UI tab is added when the frontend is ready.

PROJECT STATES

State	Definition	What VM can do
Draft	Project created. AI analysis complete or in progress. VM confirming sections. No actors invited yet.	Upload documents via any channel, confirm sections, edit extractions, use AI chat per section, resolve conflicts.
Active	All sections confirmed by VM. VM has explicitly activated the project. Actors can be invited.	Invite actors, upload new documents, continue iterative improvement, advance phases, calculate royalties.
Archived	Project completed or discontinued. All data preserved. No new actions possible.	Read-only access to all project data and history.

DECIDED — DO NOT REOPEN

- Three input channels: Manual Upload, Connected Sources, Paste Text — all feed the same AI pipeline
- Three confidence levels: Confirmed, Provisional, Conflicting — displayed on every extracted field
- Connected Sources (Outlook/Gmail/Drive) require explicit VM approval before any item enters the project
- Conflicting fields block section confirmation until VM resolves them
- VM can override any AI extraction — all overrides logged with timestamp and original value preserved
- The VM never fills in forms from scratch — all data originates from documents
- Documents can be uploaded at any time, before and after project activation
- Confirmation order: Overview → Genotype → Supply Chain → Phases → Contracts → IP Rights → To-Do
- Chronology data model is active from day one — UI tab added later
- Quarantine status is computed on every read — never stored directly
- GET /api/protocols/active endpoint must be built in MVP for NuBred Node integration

OPEN — TEAM TO IMPLEMENT

- Confidence level badges on every extracted field — green/orange/red with tooltip showing source
- Conflict resolution UI — side-by-side comparison with VM selection and archive of discarded value
- Connected Sources integration — Outlook OAuth, Gmail OAuth, Google Drive OAuth with review queue
- Paste text interface — input area with document type selector and immediate AI analysis
- AI analysis loading state — progress by section, not a spinner
- Section AI chat — contextual to current section, responds only from uploaded documents
- Why Button — present on every extracted field showing document, passage, framework element
- Chronology event generation — background process that runs on every document analysis

DO NOT

- Do not auto-import from Connected Sources without VM review queue approval
- Do not mark any Connected Sources extraction as Confirmed without VM override
- Do not allow section confirmation while Conflicting fields exist in that section
- Do not delete superseded values — archive them with their source
- Do not allow project activation without VM having confirmed all sections
- Do not call Claude or Gemini from the frontend — all AI calls through NestJS
- Do not store Quarantine status — compute it on every read

- Do not use a different data schema for NuBred Node observations vs manual Field Observer